

Scope and Limits of Darwinian Evolution Under Regime Transitions

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Theorem of Axiomatic Necessity (TNA)

Abstract

Darwinian evolution—based on random variation and natural selection—has proven remarkably successful in explaining adaptive change within stable environmental regimes. However, persistent anomalies such as the Cambrian Explosion, abrupt speciation events, rapid phenotypic shifts under extreme stress, and modern human morphological expansion suggest the existence of boundaries beyond which Darwinian mechanisms lose explanatory power.

This paper proposes that Darwinian evolution operates as a **local optimization process**, valid only within regimes of slow boundary variation and subcritical mutational throughput. When environmental or systemic boundary conditions undergo **discontinuous transitions**, evolutionary dynamics exit the Darwinian regime and enter a **necessity-driven phase**, governed by minimal coherence constraints ($F_{\min}$) rather than stochastic search.

We formalize this distinction, define operational thresholds for Darwinian applicability, and argue that regime transitions require a different generative mechanism—one based on structural necessity rather than cumulative selection.

1. Introduction

The question is not whether evolution occurs, but **what kind of evolution applies under which conditions**.

Modern evolutionary biology often treats Darwinian evolution as a universal engine capable of generating biological complexity across all temporal and environmental scales. This assumption implicitly extends gradualism beyond its domain of validity, transforming a successful local model into a global metaphysics.

This paper reframes the problem by asking a more precise question:

Under what conditions is Darwinian evolution operationally valid, and where does it provably fail?

2. Definitions and Conceptual Clarifications

To avoid semantic ambiguity, we define key terms explicitly.

2.1 Darwinian Evolution (DE)

Darwinian evolution is defined here as a process characterized by:

- Random heritable variation (mutations),
- Differential survival and reproduction (selection),
- Incremental accumulation of adaptive change over time.

This definition presupposes:

- Relative stability of boundary conditions,
- Sufficient time for selection to act,
- Mutational rates below an informational instability threshold.

2.2 Regime and Boundary Conditions

A **regime** is defined as a region of state space in which boundary conditions B vary continuously and slowly relative to the system's internal response times.

A **regime transition** occurs when:

$$B(t) \rightarrow B'(t) \quad \text{discontinuously or faster than adaptive relaxation}$$

Examples include:

- Abrupt climate shifts,
- Oxygenation events,
- Radiation spikes,
- Ecological collapses,

- Technological or energetic discontinuities.
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3. Darwinian Evolution as a Local Approximation

Darwinian evolution functions as a **local optimizer** in configuration space.

Formally, let:

- $\Omega(B)$ be the space of viable phenotypes under boundary conditions B ,
- Ψ be the system's mutational–selection throughput.

Darwinian dynamics approximate gradient descent on fitness landscapes **only if**:

$$\left| \frac{dB}{dt} \right| \ll \Psi_{\text{crit}}$$

Where Ψ_{crit} is the maximum mutation–selection rate before informational coherence collapses.

Within this domain, Darwinian evolution is effective, predictive, and empirically validated.

4. The Mutation Throughput Threshold

There exists a **critical mutation rate per unit time** beyond which Darwinian optimization becomes unstable.

Above this threshold:

- Selection cannot retain information,
- Functional intermediates are lost faster than they are selected,
- The system enters an error catastrophe (Eigen threshold).

Formally:

$$\Psi > \Psi_{\text{crit}} \Rightarrow \text{Selection fails to preserve coherence}$$

This establishes an **upper operational bound** for Darwinian evolution.

5. Regime Transitions and the Failure of Gradualism

During regime transitions:

- Boundary conditions shift faster than Darwinian correction can operate,
- The probability of successful stochastic search collapses,
- Intermediate forms are nonviable.

This is empirically observed in:

- The Cambrian Explosion,
- Rapid speciation after mass extinctions,
- Extremophile emergence,
- Sudden phenotypic restructuring under stress.

Darwinian evolution does not *cause* these transitions; it is **suspended by them**.

6. Necessity-Driven Phase Transitions

We propose that regime transitions invoke a different mechanism: **structural necessity**.

Let $F_{\min}$ denote the **minimal coherence condition** required for system persistence under new boundaries B' .

Under regime transition:

- The system does not search $\Omega(B')$ randomly,
- It collapses directly into configurations satisfying $F_{\min}$,
- Survival becomes binary: coherence or extinction.

This is not adaptation by accumulation, but **crystallization under constraint**.

7. Evolution Reframed: A Two-Regime Model

Evolutionary dynamics must therefore be split into two operational regimes:

REGIME	DOMINANT MECHANISM
Stable boundaries	Darwinian selection
Boundary transitions	Necessity-driven phase change

Darwinian evolution optimizes *within* regimes.

Necessity determines *which regimes exist*.

8. Implications for Evolutionary Theory

This framework:

- Preserves Darwinian evolution where it works,
- Explains its systematic failures without ad hoc extensions,
- Eliminates the need for probabilistic miracles,
- Aligns biological evolution with thermodynamics and information theory.

Attempts such as the “Extended Evolutionary Synthesis” remain descriptive because they do not introduce a generative mechanism for regime formation.

9. Conclusion

Darwinian evolution is not false.

It is **bounded**.

It explains variation within regimes but not the origin of regimes themselves. When boundary conditions shift discontinuously, selection gives way to structure, and probability yields to necessity.

Trying to explain regime-scale biological transitions using gradualist fossil accumulation is akin to arranging candles and expecting them to generate lightning.

The lightning comes from the boundary conditions.

Acknowledgments

The author thanks the broader interdisciplinary literature in thermodynamics, information theory, and systems physics for providing the tools necessary to reframe evolutionary dynamics beyond metaphor.

Appendix A — Formal Bifurcation Mathematics of Regime Transitions

A.1 System Definition

Let a biological system be represented by a state vector

$$x(t) \in \mathbb{R}^n$$

describing phenotypic, metabolic, and regulatory variables.

The system evolves under boundary conditions

$$B(t) \in \mathbb{R}^m$$

(environmental, energetic, chemical, climatic constraints).

The general dynamics are:

$$\frac{dx}{dt} = F(x, B)$$

where F encodes internal biological processes (development, metabolism, reproduction).

A.2 Coherence Functional and Viability Domain

Define a **coherence functional**

$$\mathcal{C}(x, B) : \mathbb{R}^n \times \mathbb{R}^m \rightarrow \mathbb{R}$$

such that:

$$\mathcal{C}(x, B) \geq 0 \quad (\text{viable state})$$

The **viability domain** under boundary conditions B is:

$$\Omega(B) = \{x \mid \mathcal{C}(x, B) \geq 0\}$$

A.3 Minimal Coherence Constraint $F_{\min}$

Define the **minimal coherence condition**:

$$F_{\min}(B) = \inf_{x \in \Omega(B)} \mathcal{C}(x, B)$$

Persistence of the system requires:

$$F_{\min}(B) \geq 0$$

When boundary conditions shift, the system must satisfy:

$$\exists x' \in \Omega(B') \text{ such that } \mathcal{C}(x', B') \geq 0$$

Otherwise, extinction is unavoidable.

A.4 Darwinian Regime (Continuous Boundaries)

Assume:

$$\left| \frac{dB}{dt} \right| \ll \tau^{-1}$$

where τ is the internal relaxation time of the system.

Under this condition:

- $\Omega(B)$ deforms smoothly,
- local gradients exist,
- Darwinian selection approximates gradient descent on fitness/coherence landscapes.

Formally:

$$x(t) \approx x^*(B(t))$$

where x^* is a local attractor.

A.5 Bifurcation Condition (Regime Transition)

A **regime transition** occurs when:

$$\left| \frac{dB}{dt} \right| \geq \tau^{-1} \quad \text{or} \quad \Omega(B) \cap \Omega(B') = \emptyset$$

At this point:

- continuous trajectories $x(t)$ cease to exist,
- no path of small variations connects viable states,
- Darwinian dynamics break down.

This is a **topological bifurcation** of the viability domain.

A.6 Collapse of Probabilistic Search

Let Ψ be the mutational–selection throughput (informational exploration rate).

The probability that stochastic variation finds a viable configuration after a boundary jump satisfies:

$$P_{\text{Darwin}} \sim \frac{|\Omega(B')|}{|\mathcal{X}|}$$

where $\mathcal{X}$ is the total configuration space.

For abrupt transitions:

$$|\Omega(B')| \ll |\mathcal{X}| \quad \Rightarrow \quad P_{\text{Darwin}} \rightarrow 0$$

This defines the **probability collapse**.

A.7 Necessity-Driven Transition

When Darwinian paths vanish, system persistence reduces to:

$$x' = \arg \min_x [\mathcal{C}(x, B')] \quad \text{subject to} \quad \mathcal{C}(x, B') \geq 0$$

This is **not optimization over time**, but **instantaneous structural selection** imposed by boundary constraints.

The transition is:

- discontinuous,
 - non-computable from prior states,
 - independent of gradual accumulation.
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A.8 Interpretation

Darwinian evolution corresponds to **continuous attractor tracking**.

TNA regime transitions correspond to **coherence bifurcations**.

Mathematically:

- Evolution = local dynamics on a stable manifold,
- Speciation under regime change = jump between disjoint manifolds.

No amount of stochastic sampling can bridge disjoint viability domains.

A.9 Corollary (Cambrian Case)

If:

$$\Omega(B_{\text{pre}}) \cap \Omega(B_{\text{Cambrian}}) = \emptyset$$

then:

- intermediate forms are non-viable,
- fossil continuity is impossible in principle,
- abrupt appearance is mandatory.

This is a **mathematical consequence**, not a fossil artifact.

A.10 Summary

Bifurcation mathematics shows that:

- Darwinian evolution is valid only under continuous boundary deformation,
- regime transitions enforce discontinuous jumps,
- survival becomes a problem of **existence**, not probability.

Evolution optimizes **within** regimes.

Necessity selects **between** regimes.